Limits of quadratic phase surrogates for finite-core collection in apodized beams

Auto BeamScientist Project

Abstract

A low-order phase approximation can preserve an input-plane statistic while changing the power collected after propagation. We test two quadratic surrogates for a radial quartic phase in Gaussian, Pearcey and butterfly fields: one matches the intensity-weighted mean phase, and the other minimizes the intensity-weighted squared phase-gradient error. Both coefficients are calculated before propagation without fitting output data. The observable is the median power fraction inside a fixed on-axis circular core over a free-space axial interval. Across three quartic coefficients, the first surrogate gives residual ratios of 0.81–0.89 relative to its own departure from the unmodified field. The Gaussian reference therefore fails the original proposed equivalence test. The phase-gradient surrogate also underestimates the quartic core fraction, by 9.51–23.27% across the primary cases. Complete-arm mesh, axial-grid and independent Fresnel checks distinguish a persistent surrogate discrepancy from circular-core discretization. The resulting comparisons characterize the limits of these two low-order reductions for the specified fields and collection geometry.

1 Introduction

Finite-aperture collection depends on the propagated spatial field, rather than on a single intensity or phase statistic at the source. This distinction matters when a higher-order wavefront is replaced by an equivalent defocus. A quadratic phase is convenient because it has a single coefficient and, for an untruncated Gaussian in paraxial propagation, preserves the Gaussian functional form. A radial quartic phase has a different spatial gradient and generally leaves that family.

We examine how accurately two input-derived quadratic coefficients reproduce the core energy of a quartic-phase field. The first was proposed by the autonomous research model and equates intensity-weighted mean phase. The second has a clearer wavefront interpretation: it minimizes the intensity-weighted squared difference between radial phase gradients. Neither coefficient is fitted to the measured core energy. Gaussian, Pearcey and butterfly inputs test whether the discrepancy is confined to a caustic-bearing field or already occurs for a Gaussian reference.

The comparison fixes input power, aperture, observation radius and propagation planes before measurement. It addresses a numerical approximation question; it does not optimize a new focusing device. Testing both matching rules is essential because rejecting an arbitrary surrogate alone would say little about the usefulness of quadratic reductions.

2 Related Work

Pearcey-beam studies connect the cusp diffraction integral with autofocusing and self-healing [2]. Higher-order optical catastrophe fields include selected sections of the butterfly integral [5]. Symmetric Pearcey–Gaussian beams and masked Pearcey arrays show how changes to the seed and aperture modify focusing behavior [4, 3]. These works supply structured test fields and physical context for a finite collection region. Here the field definitions remain fixed while a radial phase is replaced by a lower-order surrogate.

The numerical propagation question is separate from the source construction. Band-limited angular-spectrum methods address sampling of the free-space propagator [1]. Our calculation uses a conservative radial cutoff and checks selected fields with a spatial Fresnel integral. The comparison concerns the measured discrepancy between specified phase approximations, rather than a new propagation algorithm or a new beam family.

3 Model and numerical methods

We use $U_G = \exp(-r^2/w^2)$ and $U_n = C_n(x/s, y/s)\exp(-r^2/w^2)$ inside a circular aperture, where

$$C_n(u,v) = \int_{\mathbb{R}} e^{i(t^n + vt^2 + ut)}\, dt, \qquad n = 4 \text{ (Pearcey)}, \quad n = 6 \text{ (butterfly section)}. \tag{1}$$

The unused higher butterfly control coefficients are zero. The integrals use a rotated contour, with the generator's established 192/384-point consistency check. Every sampled input is normalized by

$$U_0 = \sqrt{\frac{P}{\Delta x^2 \sum_{j,k} |V_{jk}|^2}}V, \quad U_4 = U_0 e^{ia_4\rho^4}, \quad U_2 = U_0 e^{ia_2\rho^2}, \quad \rho = r/w. \tag{2}$$

The moment surrogate is the particular proposal

$$M = \frac{\sum |U_0|^2 \rho^4}{\sum |U_0|^2 \rho^2}, \qquad a_2 = a_4 M. \tag{3}$$

It is calculated on the incident field, without fitting propagated outcomes. It is not asserted to be an exact defocus equivalence. Its phase gradient is linear in radius, whereas the quartic gradient is cubic.

Homogeneous free-space propagation uses the scalar angular-spectrum operator

$$U_z = \mathcal{F}^{-1}\left[\mathcal{F}(U_{\text{in}}) B_z e^{i2\pi z\sqrt{\lambda^{-2} - f_x^2 - f_y^2}}\right], \quad B_z = \mathbf{1}_{f_x^2 + f_y^2 \le \min(\lambda^{-2}, f_c^2)}, \quad f_c = \frac{\lambda^{-1}}{\sqrt{1 + (2z/L)^2}}. \tag{4}$$

This radial cutoff is conservative and differs from the full Matsushima passband. There is no ideal lens. All fields use $P = 1$ W, $\lambda = 633$ nm, $w = 300$ μm, aperture radius 0.9 mm and catastrophe scale $s = 200$ μm. The primary grid is 384^2 over $L = 2.4$ mm. Each case has 13 uniformly spaced planes over 20–80 mm, core radius $r_c = 50$ μm, and

$$C(z) = P^{-1}\int_{r \le r_c} |U_z|^2 dA, \qquad \eta = \text{median}_{z \in \mathcal{Z}}\, C(z), \qquad R = \frac{|\eta_4 - \eta_2|}{|\eta_2 - \eta_0|}. \tag{5}$$

Circular integration uses 32×32 subpixel samples only for boundary cells; the original pixel-mask values are also saved. These supplementary values are kept separate from the earlier $N = 192$ pixel-mask outcomes. Phase coefficients are $a_4 = 0.3, 0.6, 0.7$; every baseline and surrogate shares the same power, aperture, grid and objective. The original Gaussian anchor is $R \le 0.15$ at $a_4 = 0.7$. The higher-coefficient scan is a post-exploration follow-up; it does not redefine that threshold.

All arms pass retained spectral power ≥ 0.995, boundary energy ≤ 0.01, and input-power error $< 10^{-8}$. Additional checks recompute all three arms for Gaussian and Pearcey at $a_4 = 0.7$: a 768^2 mesh, independent separable real-space Fresnel quadrature at 768^2, and a true 41-plane axial grid at 384^2. These are fixed-field checks without refitting. Butterfly has the primary scan only. Airy is excluded because its original sampling/boundary checks failed.

Field	a_4	a_2	η_0	η_4	η_2	R
Gaussian	0.3	0.30000	0.053382	0.053020	0.050096	0.8897
Gaussian	0.6	0.60000	0.053382	0.052645	0.047101	0.8827
Gaussian	0.7	0.70000	0.053382	0.052518	0.046160	0.8803
Pearcey	0.3	0.26657	0.061577	0.060996	0.058113	0.8321
Pearcey	0.6	0.53314	0.061577	0.060318	0.054929	0.8106
Pearcey	0.7	0.62200	0.061577	0.060204	0.053925	0.8205
Butterfly	0.3	0.25836	0.063678	0.063190	0.060346	0.8536
Butterfly	0.6	0.51671	0.063678	0.062779	0.057179	0.8616
Butterfly	0.7	0.60283	0.063678	0.062609	0.056177	0.8574

Table 1: Primary fixed-core comparison. All three propagated arms are measured directly. Values are fractions of the same 1 W input; they are not peak-intensity gains.

Field	Check	Planes	η_0	η_4	η_2	R
Gaussian	ASM 768	13	0.053390	0.052528	0.046167	0.8806
Gaussian	Fresnel 768	13	0.053390	0.052528	0.046167	0.8806
Gaussian	ASM 384	41	0.053382	0.052518	0.046160	0.8803
Pearcey	ASM 768	13	0.061587	0.060216	0.053933	0.8208
Pearcey	Fresnel 768	13	0.061587	0.060216	0.053933	0.8208
Pearcey	ASM 384	41	0.061577	0.060226	0.053925	0.8235

Table 2: Complete-arm confirmation at $a_4 = 0.7$. Mesh and direct-Fresnel checks use 13 planes; the axial check uses 41 actual planes. All checks retain the original physical window. The field identities are explicit.

4 Results

The primary residuals range from 0.8106 to 0.8897. The Gaussian anchor fails, and the caustic residuals do not exceed the Gaussian residual. The quartic and surrogate core fractions differ by 4.50–12.11% relative to the quartic fraction, with the minimum at butterfly $a_4 = 0.3$ and the maximum at Gaussian $a_4 = 0.7$. This is a counterexample to the tested surrogate for this geometry; it does not identify a caustic-specific mechanism.

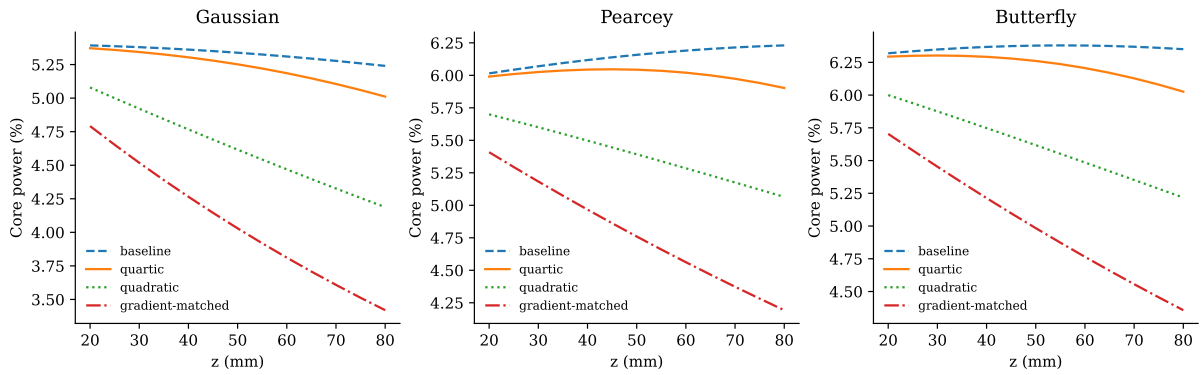


Figure 1: Free-space core curves for the unmodified field, quartic phase a4=0.7, and mean-phase-matched quadratic and least-squares phase-gradient-matched quadratic phases. All arms use 1 W input, a 0.9 mm aperture, N=384 and a 50 micrometre collection radius; no ideal lens is applied.

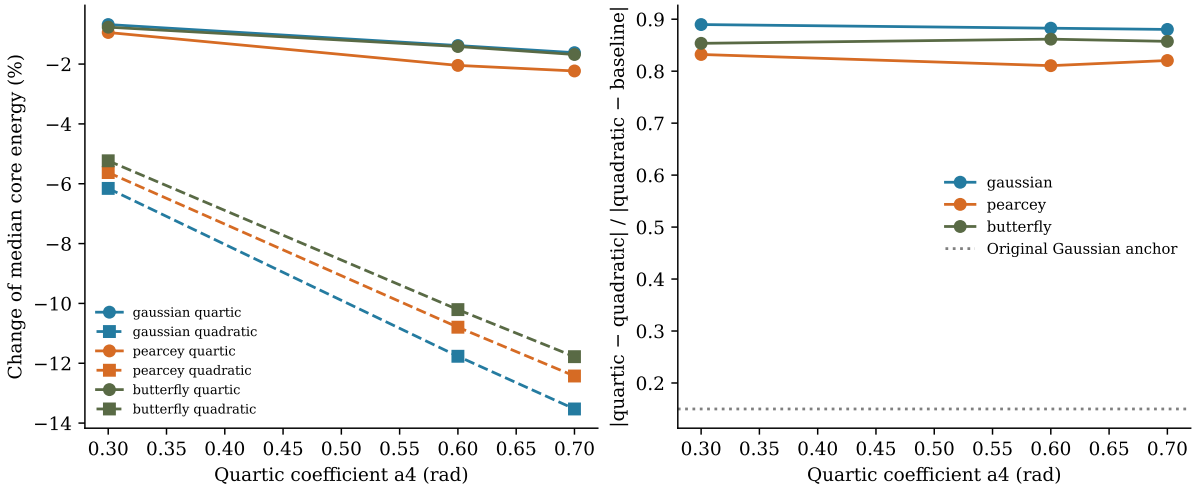


Figure 2: Moment-matched quadratic comparison at three fixed quartic coefficients. The right panel measures residual relative to the quadratic response, rather than declaring two visually similar patterns equivalent. The 15% line is the original Gaussian equivalence anchor; it is not a universal acceptance threshold for every family.

Field	a_4	$a_{2,\nabla}$	$\eta_{2,\nabla}$
Gaussian	0.3	0.60000	0.047101
Gaussian	0.6	1.20000	0.041855
Gaussian	0.7	1.40000	0.040298
Pearcey	0.3	0.53314	0.054929
Pearcey	0.6	1.06629	0.049290
Pearcey	0.7	1.24400	0.047599
Butterfly	0.3	0.51671	0.057179
Butterfly	0.6	1.03342	0.051538
Butterfly	0.7	1.20566	0.049840

Table 3: Direct propagation of the least-squares phase-gradient surrogate on the same primary grid. Compare these core fractions with the quartic and base columns of Table 1.

4.1 A phase-gradient-matched quadratic control

The first surrogate matches mean phase, since $a_2\langle\rho^2\rangle_I = a_4\langle\rho^4\rangle_I$. A second reviewed control instead minimizes

$$E(a_2) = \int |U_0|^2 \left|4a_4 r^3/w^4 - 2a_2 r/w^2\right|^2 dA, \qquad a_{2,\nabla} = 2a_4 \frac{\langle\rho^4\rangle_I}{\langle\rho^2\rangle_I} = 2a_4 M. \tag{6}$$

This analytic coefficient is fixed without output fitting. The reviewed follow-up permits coefficients up to 1.4 rad for this control, beyond the generic hypothesis-interface chirp bound; it does not change the recorded primary cases.

5 Discussion

The first matching rule equates $\langle\phi_2\rangle_I$ and $\langle\phi_4\rangle_I$. Equality of mean phase does not imply equality of a wavefront: a constant phase itself has no intensity consequence, while the remaining spatial phase determines interference after propagation. All phase-only arms have exactly the same

incident intensity and therefore exactly the same incident intensity moments. Their propagated core fractions can nevertheless differ.

The gradient-matched coefficient removes a stronger ambiguity. It is the least-squares quadratic approximation to the quartic radial phase gradient under the frozen incident-intensity weighting. Its result is reported as a separate propagated arm. Matching that weighted gradient still compresses an entire spatial function into one number. The core integral samples the propagated field inside a finite region, and it need not preserve the ordering or error measure of the source-plane fit. In particular, a Gaussian multiplied by a quadratic phase stays Gaussian in the ideal untruncated paraxial model; multiplication by a quartic phase does not. A claim that the Gaussian q parameter exhausts the quartic response is therefore incorrect.

The original Gaussian equivalence anchor fails before any family-specific interpretation is considered. The caustic fields also show a substantial discrepancy, but their residuals are not systematically greater than the Gaussian residual. That observation prevents the proposed residual from isolating a caustic-specific effect. The complete-arm checks retain field identity, propagation planes and matching coefficients, so comparisons between different families are not mistaken for convergence errors.

These results apply to a fixed core and axial window. A surrogate optimized directly for this output functional, or a different collection radius, could behave differently. Such a fit would answer a different question from the input-derived, zero-output-fit comparisons performed here.

6 Conclusion

For the tested finite-aperture Gaussian and catastrophe fields, equating intensity-weighted mean phase does not reproduce the median core energy produced by a quartic phase. The original Gaussian anchor fails with a residual ratio near 0.88, and the caustic residuals do not provide a separate family-specific signature. The additional least-squares gradient control underestimates quartic core fractions by 9.51–23.27%, so its better-defined source-plane fit does not improve this output prediction. The tables and saved axial curves report both comparisons under common power and sampling, with complete-arm numerical checks for the selected Gaussian and Pearcey cases. The supported result is a measured limit of these specified phase surrogates for finite-core prediction.

Code and acknowledgement

Auto BeamScientist adapts Sakana AI Scientist-v2. Code and an offline reproduction script are available at `https://github.com/yurunsheng1/beamscientist`.

References

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